

MTGA Log Parser — System Diagram

Single-pass, event-driven pipeline · zero runtime dependencies · ESM-only TypeScript

INPUT

UTC_Log - *.log files
MTGA client output, sorted by date

ENTRY · src/logParser.ts

parseAllLogsWithDebug(config)
read dir · split lines · loop files

parseLinesAndAddToSession
single-pass scan · 3 sentinel substrings



PER-LINE HANDLERS

handleParseDeck
deckParser.ts

extractDeckInfo
applyCoursesPayload
→ deckByEvent / pendingDeck

handleParseMatchStateChange
matchHandler.ts

handleMatchStart (filtered)
handleMatchEnd
→ matchMap · seat↔team map

handleParseGREEventLine
logParser.ts

corrects localTeamId via seat
tryExtractGameState
→ fans out to collectors

SUB-COLLECTORS · per GameStateMessage

rawGameObjects.ts
GRE JSON → typed structs

toGameObject · mergeGameObject
toZone · toPlayer · toBoardCard

two-pass annotation loop builds GameActions

gameDataCollector
gameDataParser.ts

mulligans
final life totals
turn count
end reason

boardStateCollector
boardStateParser.ts

phase snapshots
turn draw records
game actions
rawState (debug)

ACCUMULATED SESSION STATE · Maps passed by reference

Session
matchMap · localTeamIdMap · opponentGrpIds · deckByEvent · deckUsages
gameEndReasonsMap · seatToTeamByMatch · pendingDeckName/List · currentMatchId

POST-PROCESSING · after all files parsed

resolveColors (parallel)
Promise.all over opponentGrpIds
optional callback → match.opponentColors

patch gameEndReason
from finalMatchResult.resultList
life · concede · timeout · draw · unknown

OUTPUT

ParseResult
matches · gameSnapshots · boardSnapshots · turnDrawRecords · gameActions
myDeckListMap · deckUsages · opponentGrpIds · debugBoardState(matchId, gameNumber)